

Problem 1.7

Compounding is the process of adding accumulated interest back to the principal, so that interest is earned on interest from that moment on. In this case, we have the formula

$$A = P(1 + i)^t$$

and we call it a yearly compound interest. You can think of compound interest as a series of back-to-back simple interest contracts. The interest earned in each period is added to the principal of the previous period to become the principal for the next period.

You borrow \$10,000 for three years at 5% annual interest compounded annually. What is the amount value at the end of three years?

Solution

Using the compound interest formula

$$A = P(1 + i)^t,$$

with

$$P = 10,000, \quad i = 0.05, \quad t = 3,$$

we get

$$A = 10,000(1 + 0.05)^3.$$

Therefore,

$$A = 10,000(1.05)^3 = 10,000(1.157625) = 11,576.25.$$

Hence, the amount value at the end of three years is

$$\boxed{\$11,576.25}.$$